Microplastic Waste in Agriculture: An Emerging Threat to Soil, Food, and Human Health

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Plastics have revolutionized modern life, much like the industrial revolution itself. Their widespread use has transformed packaging, construction, transportation, medicine, electronics, and agriculture. Today, nearly 450 million tonnes of plastic are produced globally each year, and approximately 12.5 million tonnes are used in agriculture alone (*Bergmann et al.*, 2022; Hofmann *et al.*, 2023). While plastics have improved agricultural efficiency and crop productivity, their long-term environmental consequences are becoming increasingly evident.

The Growing Global Plastic Crisis

Plastic waste generation has increased dramatically and is now recognized as one of the most urgent environmental challenges of our time. According to projections by the Organisation for Economic Co-operation and Development (OECD), global plastic waste is expected to nearly triple from 353 million metric tonnes in 2019 to about 1,014 million metric tonnes by 2060. Alarmingly, only 10% of plastics are recycled, while 14% are incinerated. The remaining 76% accumulate in landfills or leak into the environment, posing serious threats to ecosystems and human health. One of the most concerning outcomes of this plastic overload is the formation of microplastics plastic particles smaller than 5 mm and nano plastics, which are less than 1 μm in size. Due to their persistence, small size, and mobility, these particles are now found in nearly all environmental compartments, including agricultural soils.

Plasticulture and Agriculture's Dependence on Plastics

The extensive use of plastics in farming has given rise to the term "plasticulture," which refers to the application of plastic materials to enhance agricultural efficiency and crop yield (Sahai *et al.*, 2025). The development of polyethylene polymers in the late 1930s laid the foundation for plasticulture, which expanded rapidly during the 1950s and 1960s with the introduction of plastic mulches, greenhouse films, drip irrigation tubes, silage wraps, and fertilizer coatings.

Today, plastics are widely used to regulate soil temperature, conserve water, suppress weeds, and improve crop quality. Demand for agricultural plastics is projected to increase by 50% by 2030, driven by climate change and the need for higher food production (FAO, 2021). However, these benefits come at a significant environmental cost.

Sources of Microplastics in Agricultural Soils

Agricultural soils have become major reservoirs of microplastics, receiving contamination through multiple pathways.

The most important sources include:

- Plastic mulching films, greenhouse covers, and irrigation systems that fragment over time due to UV radiation, temperature fluctuations, mechanical abrasion and photo-oxidation.
- Sewage sludge, compost, and organic fertilizers, which contain microplastics originating from household waste, synthetic textiles and personal care products.

- Fertilizer coatings, particularly slow- and controlled-release inorganic fertilizers made from non-biodegradable polymers.
- Irrigation water, contaminated rainwater, surface runoff, and atmospheric deposition.
- Farm activities, including littering, degradation of plastic tools, packaging materials, and animal feed plastics.

Once introduced, plastics degrade very slowly. Even so-called biodegradable plastics-such as polylactic acid (PLA)-do not always fully mineralize and can still generate microplastic residues (Hofmann *et al*., 2023).

Types of Plastics Found in Agricultural Soils
Microplastics in agricultural soils originate from a wide range of polymers.
These include:
Non-biodegradable plastics such as polyethylene (PE), polypropylene (PP), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polystyrene (PS), polyester (PES), polyamide (PA), and polyacrylonitrile (PAN).
Biodegradable plastics (BP) derived from natural polymers like starch, cellulose, and fatty acids, or modified conventional plastics containing biodegradable additives.

The environmental behavior and toxicity of these plastics depend on their polymer type, size, shape, aging, additives and exposure time.

Impacts of Microplastics on Soil Health and Ecosystems
Microplastics pose serious risks to soil structure and function. Their accumulation can alter soil porosity, aggregation, and water-holding capacity, ultimately affecting root growth and nutrient availability. Microplastics also disrupt soil microbial communities, including bacteria and fungi responsible for nutrient cycling and organic matter decomposition, thereby reducing long-term soil fertility. In addition, microplastics act as carriers for pesticides, heavy metals, and other toxic chemicals, increasing their persistence in soils. Many plastics contain additives such as plasticizers, stabilizers, flame retardants, and pigments-some of which are known endocrine disruptors that negatively affect living organisms.

Soil fauna are also affected. Earthworms and other soil organisms have been shown to ingest microplastics, leading to physical harm, reduced growth, and altered behavior. As microplastics move through soil food webs, they pose risks across multiple trophic levels.

Microplastics, Food Safety and Human Health
Emerging evidence suggests that very small microplastics and nanoplastics can be taken up by plant roots and translocated to edible plant parts. This raises serious concerns about food safety and human exposure, particularly as microplastics accumulate in staple crops over time.
Agricultural soils are not only victims of contamination but also sources of microplastics to surrounding environments. Wind erosion, runoff, and drainage transport plastic particles into rivers, lakes, and oceans, linking terrestrial agriculture to global freshwater and marine pollution.

Addressing the Microplastic Challenge in Agriculture
Microplastic pollution in agriculture is considered poorly reversible and represents a long-term threat to planetary and human health. Addressing this issue requires a combination of preventive and management strategies, including:

- Reducing reliance on conventional plastic materials in farming.
- Developing and adopting truly biodegradable and sustainable alternatives.
- Improving plastic collection, recycling, and waste management systems.
- Regulating the quality of organic fertilizers and sewage sludge applied to farmland.
- Raising farmer awareness and strengthening environmental policies.
- Supporting continued research on microplastic behavior, impacts, and mitigation in soil systems.

Conclusion

While plastics have undeniably improved agricultural productivity, their unchecked use has created a silent but persistent pollution problem beneath our feet. Microplastics in agricultural soils threaten soil health, crop productivity, ecosystem stability, and food safety. As global plastic use continues to rise, urgent action is needed to balance agricultural benefits with environmental protection. Safeguarding soil-the foundation of food systems-will be critical to ensuring sustainable agriculture and protecting human health in the decades to come

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